



Vendor · Canonical Shape Directory

Reference · v8 · case-encoded across 4 dims

The 5-segment vendor identity — locked grammar. $\text{PAY_SEQ} \times \text{SELLS} \times \text{OPM} \times \text{PMS} \times \text{IDENTITY}$. All four architectural dimensions are case-encoded (uppercase primary + optional lowercase alt). Brand scope is still computed from the RM ring at query time.

SECTION 1

The 5-segment vendor identity

The vendor canonical code carries five locked dimensions: **PAY_SEQ** (cash-flow direction), **SELLS** (what they stock — RM scope), **OPM** (how we order), **PMS** (which payment methods they accept), and **IDENTITY** (unique abbreviation). Brand scope is still a **computed view** derived from the RM ring at query time.

v8 doctrine fix: all four architectural dimensions follow the same case-encoded pattern — **uppercase letter = primary rail/direction**, **lowercase letter = also possible (alternate)**, **letter absent = impossible**. The grammar fully encodes possibility from impossibility — no information is thrown away. Every dimension is enumerated; the constraint set rejects impossible combinations.

TOTAL VENDORS

—

live in editor

PAY-FIRST PRIMARY

—

Pf + Pfr

RECEIVE-FIRST PRIMARY

—

Rf + Rfp

LINKED VS ORPHAN

—

RM-referenced vendors

[↗ Vendor Editor](#)

[↗ RM Editor](#)

[↗ RM Shape Directory](#)

[↗ COA Doctrine](#)

[↓ Live Census](#)

SECTION 2

Code Anatomy — 5 segments, left to right

**{PAY_SEQ} - {SELLS} - {OPM} - {PMS} -
{IDENTITY}**

SELLS

POSITION 2 · 1-2 CHARS · WHAT THEY STOCK (RM SCOPE)

L = loose only · **B** = branded only · **Lb** = loose primary + branded alt · **Bl** = branded primary + loose alt. Flat LB retired in v8.

4 values · L · B · Lb · Bl

PMS

POSITION 4 · 1-2 CHARS · PAYMENT-METHOD SUPPORT

C = cash only · **B** = bank only · **Cb** = cash primary + bank alt · **Bc** = bank primary + cash alt.

4 values · C · B · Cb · Bc

IDENTITY

POSITION 5 · 3-10 CHARS UPPERCASE

Unique abbreviation drawn from the vendor name. Auto-derived (consonants vowel-stripped); owner-overridable. Globally unique across all vendors regardless of the four leading dimensions.

PRABHU · ZEPTO · SHARIFF · HYPERPURE · WALKIN

Reading order: PAY_SEQ (cash direction) → SELLS (what they stock) → OPM (how we order) → PMS (how we pay) → IDENTITY (who they are). Five locked dimensions, all enumerated. **Universal case convention:** uppercase letter = primary, lowercase letter = also possible, letter absent = impossible. Brand scope still computed from the RM ring upstream.

SECTION 3

PAY_SEQ directory Position 1 · 2-3 chars

Cash-flow direction. v8: case-encoded primary + alt — same pattern that runs across SELLS, OPM, and PMS.

Letter table

CODE	MEANING	REAL EXAMPLES
Pf	Pay-first only. Money always leaves our account before goods/service arrives. Pure quick commerce, strict prepaid.	Pf-B-A-B-ZEPT0 · Pf-L-M-C-WALKIN
Rf	Receive-first only. Goods/service always arrives before money goes out. Pure daily milk, post-paid utilities, strict credit-term vendors.	Rf-L-M-C-PRABHU · Rf-Lb-M-Cb- SHARIFF
Pfr	Pay-first primary + Receive-first alt. Default flow is pay-first, but receive-first is also possible (e.g. a quick-commerce vendor that occasionally ships COD).	Pfr-B-Am-Bc- EXAMPLE (hypothetical)
Rfp	Receive-first primary + Pay-first alt. Default flow is receive-first, but pay-first happens occasionally (e.g. a credit-term vendor where rare items require advance payment).	Rfp-Lb-Ma-Cb- SHARIFF (hypothetical)

Case-encoded combinations · primary/alt convention

When a vendor's cash-flow direction admits both flows, the canonical code uses three letters: leading **uppercase** = primary direction; the trailing f is a legibility suffix; trailing **lowercase** letter = alternate direction. The dropped information from a flat binary — which direction dominates — is preserved here.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
PAY_SEQ outside {Pf, Rf, Pfr, Rfp}	Enumerated set of legal codes. The 'f' suffix anchors the primary letter; Pfr / Rpf are rejected.
Pf + Rf as separate vendors for the same entity	Once a real vendor admits both flows, it should be Pfr or Rfp (primary specified) — never two records.

Why not a 6-letter V-shape here?

The richer V-shapes (V1a / V1b / V2 / V3 / V4 / V5) capture the precise cash-flow choreography of a single purchase event. Those live at the *action layer* and modulate

per-purchase, per-supplier within a single vendor record. PAY_SEQ at the vendor level only captures the directional possibility space (which flows are possible, which is dominant) — that's enough for the entity ring; the action ring fills in the granularity.

SECTION 3.5

SELLS directory Position 2 · 1-2 chars

What the vendor stocks at the RM-purchase layer. v8: case-encoded primary + alt. Flat LB retired.

Letter table

CODE	MEANING	REAL EXAMPLES
L	Loose only. Vendor only deals in unbranded RM by weight/volume — daily milk, vegetables, mutton, fish, walk-in emergency procurement.	Rf-L-M-C-PRABHU (loose milk)
B	Branded only. Vendor only stocks SKU-coded branded goods — quick commerce apps, B2B platforms, dedicated brand suppliers.	Pf-B-A-B-ZEPT0 · Rf-B-A-Bc-HYPERPURE
Lb	Loose primary + Branded alt. Mixed shop where loose dominates (by volume/value) but branded packs are also stocked as a secondary line.	Rf-Lb-M-Cb-SHARIFF (loose-led departmental)
BL	Branded primary + Loose alt. Mixed shop where branded SKUs dominate but some loose staples also stocked.	Rf-BL-M-Cb-NILDIST (hypothetical brand-led distributor)

Case-encoded combinations · primary/alt convention

When a vendor stocks both loose and branded, the canonical code uses two letters: **uppercase** = primary stock type (the bulk of what they sell); **lowercase** = alternate (the secondary line). The information that was discarded by the flat LB alphabet is preserved here.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
SELLS = LB (flat alphabetical)	v8 rejects flat ordering — owner must declare a primary letter. Use Lb or BL.
SELLS empty	Every RM-supplying vendor stocks at least one of {L, B}. Empty SELLS would be an unsupplied vendor — not a

FORBIDDEN

WHY CAUGHT

valid entity-ring state.

Lowercase first (lB) / both upper (LB) / duplicates (LL)

Same constraints as PMS. Primary is uppercase; alt is lowercase; each letter at most once.

Letter outside {L, B}

The two-class enumeration is exhaustive in RM-purchase scope. New letters S/C reserved for the bill/expense scope.

Future SELLS extension

When the bill/expense scope opens (rent, utilities, services, capex), the SELLS alphabet extends to include S (services) and C (capex). New combinations like Sb, Cl, S become valid points in the same case-encoded space — **no migration required**. The constraint set widens; the existing {L, B, Lb, Bl} rows stay valid.

SECTION 3.6

OPM directory Position 3 · 1-2 chars

Order-Placement Method — how an order with this vendor is initiated. v8: case-encoded primary + alt. Same pattern as SELLS / PMS.

Letter table

CODE	MEANING	REAL EXAMPLES
M	Manual only. Ordering always routes through a human channel — phone call, WhatsApp, walk-in, or a fixed daily/weekly delivery route. No app/API in the loop.	Rf-L-M-C-PRABHU (daily route) · Rf-Lb-M-Cb-SHARIFF (WA)
A	Automatable only. Ordering always goes via app, web portal, or API — quick commerce, B2B platforms, marketplace integrations.	Pf-B-A-B-ZEPT0 (app) · Rf-B-A-Bc-HYPERPURE (B2B)
Ma	Manual primary + Automatable alt. Default ordering is manual but the vendor also exposes an app/API as a backup channel.	Rf-Lb-Ma-Cb-EXAMPLE (hypothetical hybrid)
Am	Automatable primary + Manual alt. Default ordering is via app/API but a human channel exists as fallback (call vendor when API down).	Pf-B-Am-B-EXAMPLEAPP (hypothetical)

Case-encoded combinations · primary/alt convention

When a vendor admits both ordering channels, the canonical code uses two letters: **uppercase** = primary method (the channel that handles most orders); **lowercase** = alternate (the backup channel). The primary letter is what gates whether agents emit orders by default.

Forbidden combinations

FORBIDDEN	WHY CAUGHT
OPM outside {M, A, Ma, Am}	Same case-encoded pattern as SELLS / PMS.
Lowercase first (mA) / both upper (MA) / duplicates (Mm)	Primary uppercase, alt lowercase, each letter at most once.
Letter outside {M, A}	The two-method enumeration is

FORBIDDEN

WHY CAUGHT

exhaustive at the entity layer. New ordering channels would require a doctrine extension.

Why this dimension matters

OPM separates "needs a human to send WA" from "Cron Worker can place the order." The primary letter is the gate that decides whether an action becomes agent-emitted (A / Am) or staff-emitted (M / Ma) *by default*. The alt letter declares the fallback path is also available — useful for graceful degradation when the primary channel fails.

SECTION 3.7

PMS directory Position 4 · 1-2 chars

Payment-Method Support — which payment rails this vendor accepts. Two letters, case-encoded for hybrid primaries. Mirrors SOURCING Lb/Bl and USAGE Pr/Rp.

Letter set · pure rails

CODE	MEANING	REAL EXAMPLES
C	Cash-only. Vendor doesn't accept bank rails — no UPI, no NEFT, no card. Settlement always physical.	Rf-L-M-C-PRABHU (small loose milk vendor)
B	Bank-only. Vendor doesn't accept cash — typical for app-based platforms, registered B2B suppliers, online subscriptions, GST-registered wholesalers. UPI / NEFT / card / cheque only.	Pf-B-A-B-ZEPTO

Case-encoded combinations · hybrid rails

When a vendor accepts both rails, the canonical code uses two letters: **uppercase** = primary (default) rail, **lowercase** = alternate. The information thrown away by a flat H letter — which rail is the default — is preserved here.

CODE	MEANING	REAL EXAMPLES
Cb	Hybrid · Cash primary. Vendor accepts both, but cash is the default rail; bank is the alternate. Common for small/local vendors who occasionally take UPI.	Rf-Lb-M-Cb-SHARIFF (departmental store)
Bc	Hybrid · Bank primary. Vendor accepts both, but bank is the default rail; cash is the alternate. B2B platforms / registered vendors with COD option.	Rf-B-A-Bc-HYPERPURE (B2B with cash possible)

Forbidden combinations · what the regex rejects

FORBIDDEN	WHY CAUGHT
PMS empty	Every vendor accepts at least one rail. Empty PMS would mean "we cannot pay this vendor at all" — not a valid operational state.
bC · cB	Lowercase-first reverses the primary/alternate convention. Primary letter must be uppercase.
Cc · Bb	Each letter (C, B) appears at most once. Duplicate is meaningless.
CB · BC	Both uppercase declares two primaries — only one rail can be primary at a time.

FORBIDDEN

WHY CAUGHT

Letter outside {C, B}

The two-rail enumeration is exhaustive at the entity layer. New rails (e.g. crypto) would require a doctrine extension, not a free-form letter.

What PMS does NOT capture

PMS describes what the vendor *accepts* and *which rail is primary*, not what was used. The exact payment channel for a specific bill (UPI vs cash for a Cb vendor, NEFT vs UPI for a B vendor) is a property of the action — settled when the transaction happens. PMS only narrows the option space; the action picks one.

SECTION 4

IDENTITY directory Position 5 · 3-10 chars

A 3-10 character uppercase abbreviation that gives the vendor its identity within the global vendor namespace.

Length

3 to 10 characters, uppercase A-Z and 0-9 only. Codes need to stay scannable.

Source

Auto-derived from vendor name (consonants, vowels stripped, deduplicated). Owner override allowed.

Uniqueness

Globally unique across ALL vendors regardless of the four leading dimensions. Rf-
L-M-C-PRABHU blocks Pf-B-A-B-PRABHU .

Case

Always uppercase. Editor and API normalise lowercase on save.

Sample derivations

VENDOR NAME	AUTO-DERIVED IDENTITY	NOTES
Prabhu Buffalo Milk Vendor	PRABHU	Lead noun retained
Zepto	ZEPTO	Brand name as-is
Shariff Departmental Store	SHARIFF	Lead noun
Rent — Hamza HKP	RENT	Functional handle
BESCOM (Electricity)	ELECTRICITY	Function over brand
BSNL Internet	INTERNET	Function over brand

When two vendors play the same operational role (e.g. two electricity providers across outlets), use a discriminator suffix in IDENTITY — Pf-B-A-B-ELECTRICNCH, Pf-B-A-B-ELECTRICHE.

SECTION 5

Forbidden Combinations

Every anomaly the system rejects, in one place. The math literally cannot represent these.

FORBIDDEN

PAY_SEQ outside {Pf, Rf, Pfr, Rfp}

SELLS = "LB" (flat)

SELLS outside {L, B, Lb, Bl}

OPM outside {M, A, Ma, Am}

PMS outside {C, B, Cb, Bc}

Any case-encoded segment with lowercase first / both upper / duplicate le

FORBIDDEN

IDENTITY empty

IDENTITY < 3 chars

IDENTITY > 10 chars

IDENTITY collision

Lowercase / non-alphanumeric

DELETE while RM-referenced

SECTION 6

Live Vendor Census — All Vendors Today

Fetches live from `/api/vendors`. Each vendor shows its declared SELLS / OPM / PMS pills plus the brand scope computed from the RM ring. Click any code to open the editor.

Could not load live census from `/api/vendors` — Failed to fetch. The endpoint may need a valid PIN; this directory shares the editor's auth.

SECTION 7

Worked Examples — Reading codes left to right

Six codes spanning the v8 grammar: two pure-binary forms (Rf-L-M-C, Pf-B-A-B), the full-alternates form (Rfp-Lb-Ma-Cb), branded-primary hybrid (Rf-Bl-M-Bc), and intermediate cases. Each decomposed segment by segment.

1 · Rf-L-M-C-PRABHU — Daily loose milk vendor

Rf-L-M-C-PRABHU | | | | L PRABHU — identity (loose buffalo milk vendor in Shivajinagar) | | | L C — Cash-only (small vendor, no bank rail at all) | | L M — Manual order (we WhatsApp him daily) | L L — Loose only (unbranded buffalo milk by litre) L Rf — Receive-first (delivers milk daily; settled weekly/monthly)

Reads as: "Receive-first, sells loose, manually ordered, cash-only vendor named PRABHU." Computed scope from RM ring = NCH (only NCH consumes buffalo milk).

2 · Pf-B-A-B-ZEPTO — Quick commerce platform

Pf-B-A-B-ZEPTO | | | | L ZEPTO — identity (the quick commerce platform) | | | L B — Bank-only (no cash settlement; in-app UPI/card payment) | | L A — Automatable (Cron Worker can place orders via app/API) | L B — Branded only (SKU-coded packs — Zepto doesn't deal in loose) L Pf — Pay-first (paid in-app, then 10-minute delivery)

Reads as: "Pay-first, branded only, automatable, bank-only vendor named ZEPTO." Note PMS position 4 carries a single uppercase B — distinct from SELLS at position 2, even though they share the letter shape. Computed scope = HN (cross-brand — both NCH and HE order from Zepto).

3 · Rfp-Lb-Ma-Cb-SHARIFF — Full alternates: every dim carries a primary + alt

Rfp-Lb-Ma-Cb-SHARIFF | | | | L SHARIFF — identity (Shivajinagar departmental store) | | | L Cb — Cash primary + Bank alt (cash is default; UPI accepted) | | L Ma — Manual primary + Auto alt (we WA mostly; vendor's app exists as backup) | L Lb — Loose primary + Branded alt (loose is the bulk; branded packs are secondary) L Rfp — Receive-first primary + Pay-first alt (settle monthly mostly; rare items prepaid)

Reads as: "Every dimension has both rails, but one is dominant: receive-first leads cash flow, loose leads stock, manual leads ordering, cash leads payment." This is the doctrine fix in action — the v7 grammar (Rf-LB-M-Cb-SHARIFF) couldn't say which direction / stock / method dominated; v8 puts that information into the code itself. The action layer reads coordinates and routes accordingly without a lookup.

4 · Rf-B-A-Bc-HYPERPURE — B2B platform with COD option

Rf-B-A-Bc-HYPERPURE | | | | L HYPERPURE – identity (Zomato's B2B sourcing platform) | | | L Bc – Hybrid · Bank primary + Cash alt (NEFT/UPI default; cash possible on COD orders) | | L A – Automatable (B2B portal supports API) | L B – Branded only (SKU-coded wholesale) L Rf – Receive-first (delivery + invoice; settle on credit terms)

Reads as: "Receive-first, branded only, automatable, bank-primary hybrid vendor named HYPERPURE." Distinct from ZEPTO on two dimensions: PAY_SEQ (Hyperpure invoices on credit, Zepto charges in-app upfront) and PMS (Bc vs B — Hyperpure has a fall-back cash option on COD).

5 · Pf-L-M-C-WALKIN — Emergency walk-in procurement

Pf-L-M-C-WALKIN | | | | L WALKIN – identity (functional handle for ad-hoc local buys) | | | L C – Cash-only (counter purchase, immediate cash; no bank rail) | | L M – Manual order (staff walks to local shop) | L L – Loose only (vegetables / mutton emergency top-up) L Pf – Pay-first (cash handed over before goods loaded)

Reads as: "Pay-first, loose only, manually ordered, cash-only vendor named WALKIN." A functional vendor record covering all ad-hoc emergency procurement — counts as one entity even though physical shops vary.

6 · Rf-BL-M-Bc-NILDIST — Branded primary + loose alt, bank-primary hybrid

Rf-BL-M-Bc-NILDIST | | | | L NILDIST – identity (hypothetical brand-led distributor) | | | L Bc – Bank primary + Cash alt (NEFT/UPI default; cash possible on rare COD) | | L M – Manual only (call/WA – no app yet) | L BL – Branded primary + Loose alt (brand-led wholesaler with some loose staples) L Rf – Receive-first (delivery + invoice; settle on credit terms)

Reads as: "Receive-first, branded-dominant with loose alt, manual ordering only, bank-primary hybrid vendor." Demonstrates the v8 grammar's reach: SHARIFF and NILDIST share PAY_SEQ (Rf) and OPM (M), but their SELLS primary (Lb vs BL) and PMS primary (Cb vs Bc) flip — a difference that v7's flat grammar couldn't represent. The day NILDIST exposes an API, a single PUT moves the row to Rf-BL-Am-Bc-NILDIST; the canonical code re-keys; the action ring picks up the change with no migration.

SECTION 8

What stays out of the code

An obvious instinct would be to encode BRAND scope (NCH/HE/HN) directly in the vendor code: **Rf-L-M-C-NCH-PRABHU**. We deliberately don't — and the same discipline applies to several other tempting fields:

BRAND scope is derivable. It changes when the RM ring changes — without any vendor mutation. Encoding it statically would create sync drift the day NCH stops using PRABHU.

Per-transaction payment method. Whether a hybrid vendor was paid in UPI or cash for a specific bill is a property of the action, not the vendor. PMS narrows the option space; the action picks one.

Money source / L3 routing. Which bank account, which till, which UPI handle paid this vendor — that's a Layer-3 ledger concern, not a vendor identity concern.

Communication channel. Phone/WhatsApp/email is a property field. OPM captures whether ordering can be automated; the actual contact details are `data_json`.

Bank account / contact details. Pure properties — they live in `data_json`, never in the canonical code.

Five canonical dimensions; everything else is property or derived view.

In the static row

- `vendor_code` (Pf/Rf-SELLS-OPM-PMS-IDENTITY)
- `pay_seq`, `sells`, `opm`, `pms`, `identity_abbrev`, `vendor_name`
- `communication / location / payment / GST` in `data_json`

Never in the static row

- BRAND scope (derivable from RM ring)
- Per-transaction payment method (action layer)
- Money source / L3 routing (ledger)
- Communication channel (property)
- Bank account / contact (property)
- V-shape (action layer)

Future SELLS dimension extension

The current SELLS alphabet is {L, B} with case-encoded values {L, B, Lb, Bl}, valid in RM-purchase scope only. When the bill/expense scope opens (rent, utilities, services, capex), the alphabet extends: S = services, C = capex. Combinations like S, Sb, Cl become valid — same case-encoded constraint pattern, just a wider letter set.

No migration required: existing rows stay valid, the CHECK constraint widens, and new combinations slot in as new points in the same space.

SECTION 9

Connection to the RM ring + Action layer

The vendor ring is the **second entity ring**, sitting alongside the RM ring. Both feed downstream into the Action ring (purchases, productions, transfers).

Composition of an Action code (sketch)

`{ACTION}-{RM_CODE}-{VENDOR_CODE}-{V_SHAPE}-{TXN_ID}`

Examples:

PUR-NCH-AM-P-Lb-MLK-Rf-L-M-C-PRABHU-V1a-2026050801 ← daily milk
PUR-HN-AM-P-BL-BTR-Pf-B-A-B-ZEPT0-V3-2026050802 ← Amul butter
PUR-HN-AM-P-Lb-VEG-Rfp-Lb-Ma-Cb-SHARIFF-V2-2026050803 ← veg + bra

RM ring

Five segments — what was bought. Vendor refs live inside data_json at the L / B / I layers.

Vendor ring (this page)

Five segments — cash direction · what they sell · how we order · how they accept payment · who they are. Brand scope still computed via the RM ring.

Action ring

Composes RM + Vendor + V-shape into a transactional code that an Event can aggregate. Defined in COA Doctrine.

Two entity rings, one action ring, one event ring. Every action code is a coordinate; every event aggregates actions. The vendor ring stays minimal so the action ring stays composable.

[Directory](#) [COA Doctrine](#)

[Vendor Editor](#) [RM Shape](#)